

**A MINI PROJECT
ON
WOMEN SAFETY WEARABLE**



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CERTIFICATE

This is to certify that this is a bonafide work on “**WOMEN SAFETY WEARABLE**” for and has been submitted by **G. JNAPTHIKA (24B95A4709), M. JYOSHIKA (24B95A4719), U. ANUSHA (24B95A4731), V. SRI SHARAN (24B95A4732)** as a Internet of Things laboratory report, in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering, during the academic year 2024-2025. The candidate worked right under my Supervision and guidance.

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ABSTRACT

Women's safety has become a major concern in today's world, making it essential to develop smart solutions that ensure quick help during emergencies. This project presents a Women Safety Wearable Device based on IOT technology, designed to provide instant alerts in distress situations. The device integrates sensors, a GPS module, and communication technology to detect abnormal motion or manual SOS triggers. Once activated, it captures the user's real-time location and transmits it to predefined emergency contacts through SMS or a connected cloud platform. The compact and wearable design ensures portability and ease of use in daily life.

The proposed system enhances personal safety by enabling real-time monitoring and rapid response. It can send location details and alerts via GSM or Wi-Fi, making it effective even in remote areas. Additionally, optional features such as fall detection, voice activation, and cloud connectivity improve the device's reliability. By combining hardware sensors and IoT communication, this project aims to reduce response time during emergencies and empower women with a sense of security and confidence in their surroundings.

INTRODUCTION

In today's society, ensuring women's safety has become a critical challenge, as cases of harassment, assault, and violence continue to rise globally. Traditional safety measures, such as mobile applications or manual communication, often fail when victims cannot access their phones or call for help in time. Hence, there is a growing need for automated, intelligent, and easily accessible safety systems that can function independently during emergencies.

With the emergence of the Internet of Things (IoT), innovative wearable devices can now connect seamlessly with sensors, networks, and cloud services to deliver real-time data and alerts. These technologies enable the design of compact, low-power, and user-friendly wearables that ensure continuous monitoring and immediate communication.

The proposed IoT-based Women Safety Wearable Device utilizes GPS, GSM/Wi-Fi, and motion sensors to monitor the user's safety and trigger emergency alerts when abnormal movements or panic signals are detected. The system's integration of hardware and IoT platforms ensures quick data transmission to emergency contacts or authorities, regardless of location. This smart wearable not only enhances personal safety but also promotes technological empowerment and awareness in creating a safer environment for women.

Key features :

- **Real-time Tracking:** GPS module provides accurate live location.
- **Instant Alert System:** Sends emergency alerts with location via GSM/Wi-Fi.
- **Manual & Automatic Trigger:** Can be activated through SOS button or motion sensors.
- **Low Power Consumption:** Ensures longer battery life.
- **Cloud/Server Integration:** Enables remote monitoring and data storage.
- **User Safety Assurance:** Reduces response time and ensures timely help.

RESEARCH DETAILS

During our research for this project, we studied various GPS tracking and personal safety systems developed using microcontrollers and IoT technologies. Most of the earlier systems relied on Arduino boards combined with GSM and GPS modules, where the user's location was sent as an SMS to registered mobile numbers during emergencies. While this approach was functional, it had certain drawbacks such as dependency on mobile networks, higher operating costs, and limited real-time monitoring capabilities.

With the advancement of IoT, newer projects have shifted towards using ESP32 microcontrollers, which offer built-in Wi-Fi and high processing power. This allows data to be transmitted directly to a web-based interface, making tracking faster and more convenient. Our research showed that Wi-Fi-based systems are more scalable and efficient compared to GSM-based ones, especially for indoor environments.

Another key insight from our study was the concept of hybrid tracking, which uses both GPS and Wi-Fi geolocation. This ensures that the location can still be estimated even when GPS signals are weak such as indoors, tunnels, or crowded urban areas. This dual approach significantly improves the reliability of the system.

Furthermore, several safety-related research papers emphasized the inclusion of a panic or emergency button. This simple hardware input provides users with an instant way to send alerts or share their location in critical situations - a feature that can make a real difference in safety applications.

Taking inspiration from these ideas, our project integrates GPS-based location tracking, Wi-Fi fallback for geolocation, and a web dashboard for visualization. The result is a low-cost, real-time, and user-friendly tracking system that can serve multiple purposes - from personal safety and vehicle monitoring to asset tracking and smart IoT solutions.

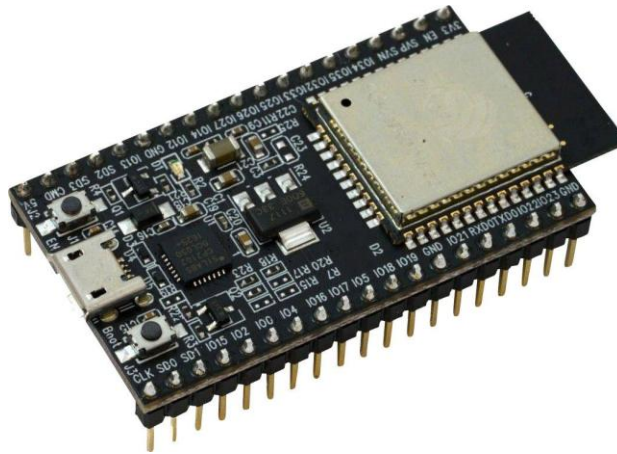
TECHNICAL STACK

SOFTWARE USED

- **ARDUINO IDE :** The Arduino Integrated Development Environment (IDE) is an open-source software used for writing, compiling, and uploading programs to microcontroller boards such as the ESP32. It provides a simple and user-friendly interface that supports C and C++ programming languages. In this project, the Arduino IDE is used to develop and upload the program that controls various modules like the GPS, Wi-Fi, and sensors. It also allows real-time monitoring and debugging of the system through the Serial Monitor, making it an essential tool for testing and verifying the working of the women safety wearable device.
- **ESP32 BOARD PACKAGES :** The ESP32 board package is an essential software component installed in the Arduino IDE to enable programming and communication with the ESP32 microcontroller. It provides the necessary board definitions, libraries, and drivers required to compile and upload code specifically for ESP32 devices. Through this package, developers can utilize the ESP32's built-in features such as Wi-Fi, Bluetooth, and multiple GPIO pins for IoT-based applications. In this project, the board package allows smooth integration of GPS, sensors, and communication modules to ensure efficient operation of the women safety wearable system.
- **LIBRARIES USED :**
 - **TinyGPS++ :** The TinyGPS++ library is used to decode and process data received from the GPS module. It extracts essential information such as latitude, longitude, speed, and time from the raw GPS signals, allowing the ESP32 to obtain the user's real-time location accurately.
 - **WiFi.h Library :** The WiFi.h library enables the ESP32 to connect to a wireless network. It handles all Wi-Fi communication functions, allowing the device to send location data or alerts to cloud platforms, web servers, or mobile devices over the internet.
 - **ArduinoJson Library :** The ArduinoJson library is used to create, parse, and manage JSON data formats. It helps in organizing sensor data, GPS coordinates, and alert messages into a structured format before transmitting them to the server or API.
 - **EEPROM.h Library:** The EEPROM.h library allows the ESP32 to store and retrieve small amounts of data permanently, even after power loss. In this project, it is used to save important information such as emergency contact numbers or configuration settings.

HARDWARE COMPONENTS

- **ESP32 MICROCONTROLLER** : The ESP32 microcontroller serves as the central processing unit of the women safety wearable system. It is a powerful and versatile board that comes with built-in Wi-Fi and Bluetooth capabilities, making it ideal for Internet of Things (IoT) applications. The ESP32 collects data from connected modules such as the GPS and accelerometer, processes this information, and sends alerts through internet-based communication. Its dual-core processor, low power consumption, and multiple GPIO pins allow it to handle several tasks simultaneously while maintaining efficiency. In this project, it manages sensor readings, triggers emergency signals, and transmits real-time location data to cloud platforms or registered contacts.



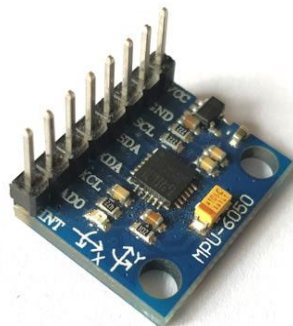
- **GPS Module (Neo-6M)** : The Neo-6M GPS module is used to obtain the geographical position of the user in terms of latitude and longitude coordinates. It communicates with the ESP32 using serial communication (TX and RX pins) and provides accurate, real-time location data. The module can receive satellite signals and continuously track the user's position even when they are in motion. In emergency situations, the ESP32 retrieves these coordinates from the GPS module and includes them in the alert message that is sent to family members or authorities. Its high sensitivity and reliability make it a crucial component for location tracking in wearable safety systems.



- **SOS Push Button / Switch** : The SOS push button is one of the most important components of the project, allowing the user to manually activate the emergency alert system. When pressed, it sends a signal to the ESP32, which then collects the user's current GPS location and sends it to the predefined emergency contacts. The button ensures that the user can immediately request help in distress situations without needing to use a mobile phone. It is designed to be easily accessible, responsive, and small enough to fit within the wearable device's compact form factor.



- **Accelerometer (MPU6050)** : The MPU6050 is a combined 3-axis accelerometer and 3-axis gyroscope sensor used to detect motion, orientation, and sudden changes in movement. In this project, it helps identify abnormal situations such as a fall, impact, or struggle. If the sensor detects a sudden or unusual motion pattern, it can automatically trigger the ESP32 to send an emergency alert without manual input. This feature is particularly useful in scenarios where the user is unable to press the SOS button. The MPU6050 communicates with the microcontroller via the I²C protocol, providing reliable motion data for smart detection.



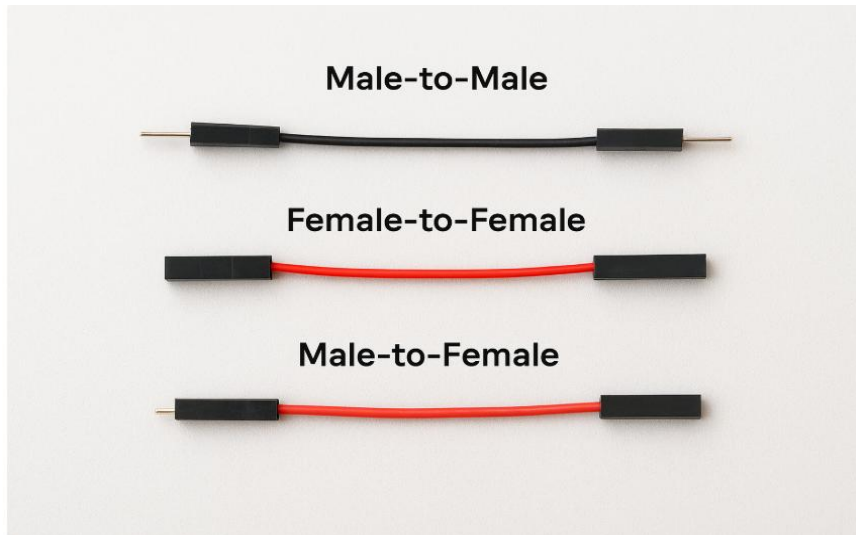
- **Buzzer** : The buzzer provides an audible alert to notify the user that the SOS signal has been successfully activated or that the system is functioning properly. It serves as an immediate feedback mechanism, confirming to the user that help is being requested. Additionally, it can also be used to attract the attention of nearby people in an emergency situation. The buzzer operates on a small voltage and is controlled through one of the GPIO pins of the ESP32, making it an efficient and simple output device.



- **LED Indicator** : The LED indicator offers a visual confirmation of the system's status. It lights up when the device is powered on, connected to Wi-Fi, or when the emergency alert is triggered. This helps users quickly verify whether the device is functioning correctly. The LED provides an easy way to monitor the operational state of the wearable system without using a separate display or mobile app, ensuring simplicity and reliability.



- **Connecting Wires** : Connecting wires are used to link all the electronic components in the circuit, enabling smooth data transfer and power distribution. They connect the ESP32 with the GPS module, accelerometer, SOS button, buzzer, and LED indicator. Proper wiring ensures stable communication between modules and prevents signal loss or loose connections during operation. The connections are made securely using jumper wires and soldered joints to maintain durability and reliability in the wearable setup.



PIN CONNECTIONS

Component	ESP32 Pin
• GPS Module (NEO-6M)	VCC → 3.3V TX → GPIO 25 RX → GPIO 4
• MPU6050	VCC → 3.3 SCL → GPIO SCL SDA → GPIO SDA
• Switch / Push Button	GPIO 33
• LED Indicator	GPIO 16
• Buzzer	GPIO 14

SAMPLE CODE

```
#include <WiFi.h>
#include <WebServer.h>
#include <HTTPClient.h>
#include <TinyGPS++.h>
#include <ArduinoJson.h>
#include <EEPROM.h>
#include <Wire.h>
#include <MPU6050.h>

// CONFIGURATION
const char* WIFI_SSID = "Srinivas-2G";
const char* WIFI_PASS = "srinu@1974";
#define EEPROM_SIZE 64

// PIN DEFINITIONS
#define GPS_RX_PIN 25
#define GPS_TX_PIN 4
#define SWITCH_PIN 33
#define LED_PIN 16
#define BUZZER_PIN 14

// OBJECTS
TinyGPSPlus gps;
HardwareSerial SerialGPS(2);
WebServer server(80);
MPU6050 mpu;

// VARIABLES
double currentLat = 0.0, currentLon = 0.0;
bool switchState = false;
bool fallDetected = false;
```

// FUNCTION DECLARATIONS

void getWiFiLocation(double &lat, double &lon);

void handleRoot();

void saveLastLocation(double lat, double lon);

void loadLastLocation();

void checkMotion(); // ☒ New function

// SETUP

void setup() {

Serial.begin(115200);

Serial.println("\n=====");

Serial.println("🚀 Booting ESP32 GPS Tracker with MPU6050...");

Serial.println("=====");

pinMode(SWITCH_PIN, INPUT_PULLUP);

pinMode(LED_PIN, OUTPUT);

pinMode(BUZZER_PIN, OUTPUT);

digitalWrite(LED_PIN, LOW);

digitalWrite(BUZZER_PIN, LOW);

EEPROM.begin(EEPROM_SIZE);

loadLastLocation();

SerialGPS.begin(9600, SERIAL_8N1, GPS_RX_PIN, GPS_TX_PIN);

//Initialize MPU6050

Wire.begin();

mpu.initialize();

if (mpu.testConnection()) {

Serial.println("✅ MPU6050 Connected Successfully!");

} else {

Serial.println("⚠️ MPU6050 Connection Failed!");

}

```

// Connect WiFi
Serial.print("📶 Connecting to WiFi");
WiFi.begin(WIFI_SSID, WIFI_PASS);
while (WiFi.status() != WL_CONNECTED) {
  delay(500);
  Serial.print(".");
}
Serial.println("\n✅ WiFi Connected!");
Serial.print("🌐 IP Address: ");
Serial.println(WiFi.localIP());

server.on("/", handleRoot);
server.begin();
Serial.println("🌐 Web Server Started Successfully!");
Serial.println("🌀 System Ready. Press the switch or shake/fall to trigger alert.\n");
}

//LOOP
void loop() {
  server.handleClient();

  // Read GPS data
  while (SerialGPS.available()) gps.encode(SerialGPS.read());
  if (gps.location.isValid()) {
    currentLat = gps.location.lat();
    currentLon = gps.location.lng();
  }

  // Check for motion/fall
  checkMotion();

  bool pressed = (digitalRead(SWITCH_PIN) == LOW);

```

```

// Continuous LED/Buzzer blink while switch held
if (pressed || fallDetected) {
    digitalWrite(LED_PIN, HIGH);
    digitalWrite(BUZZER_PIN, HIGH);
    delay(200);
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    delay(200);

    if (!switchState) {
        switchState = true;
        if (fallDetected) {
            Serial.println("\n⚠ FALL DETECTED — Sending emergency location...");
            fallDetected = false;
        } else {
            Serial.println("\n🔴 SWITCH ACTIVATED — Capturing location...");
        }

        if (!gps.location.isValid()) {
            Serial.println("⚠ GPS signal not available — using Wi-Fi location...");
            getWiFiLocation(currentLat, currentLon);
        }

        if (currentLat != 0.0 && currentLon != 0.0) {
            Serial.printf("📍 Current Coordinates: %.6f, %.6f\n", currentLat, currentLon);
            saveLastLocation(currentLat, currentLon);
            Serial.println("💾 Location saved to EEPROM memory.");
            Serial.println("🌐 Open your browser to view the map.");
        } else {
            Serial.println("❌ No valid location! Showing last saved coordinates on webpage.");
        }
    }
}

```

```

    } else {
        switchState = false;
    }
}

// MOTION DETECTION

void checkMotion() {
    int16_t ax, ay, az;
    mpu.getAcceleration(&ax, &ay, &az);

    long accelMagnitude = sqrt(ax * ax + ay * ay + az * az);

    // Simple threshold-based fall detection
    if (accelMagnitude > 25000 || accelMagnitude < 5000) {
        fallDetected = true;
        Serial.println(" 📵 Sudden Movement/Fall Detected!");
    }
}

//WEBPAGE

void handleRoot() {
    String html = "<!DOCTYPE html><html><head><meta name='viewport' content='width=device-width, initial-scale=1'>";
    html += "<meta http-equiv='refresh' content='5'>";
    html += "<title>ESP32 GPS Tracker</title>";
    html += "<style>";
    html += "body{font-family:Arial,Helvetica,sans-serif;text-align:center;background:#f9fafb;color:#222;}";
    html += "h2{color:#003366;}";
    html += "button{background:#0078D7;color:white;padding:12px 25px;border:none;border-radius:8px;font-size:16px;cursor:pointer;}";
    html += "button:hover{background:#005fa3;}";
    html += "p{font-size:16px;}";
    html += "small{color:#555;}";

```

```

html += "</style></head><body>";
html += "<h2>ESP32 Live Tracker</h2>";
if (currentLat != 0.0 && currentLon != 0.0) {
    html += "<p><b>Latitude:</b> " + String(currentLat, 6) + "<br>";
    html += "<b>Longitude:</b> " + String(currentLon, 6) + "</p>";
} else {
    html += "<p><b>No live GPS detected — showing last known location.</b></p>";
}
html += "<a href='https://www.google.com/maps?q=" + String(currentLat, 6) + "," +
String(currentLon, 6) +
    "' target='_blank'><button>View on Google Maps</button></a>";
html += "<br><br><small>Auto-refresh every 5 seconds • " + WiFi.localIP().toString() +
"</small>";
html += "</body></html>";
server.send(200, "text/html", html);
}

```

//SAVE / LOAD LOCATION

```

void saveLastLocation(double lat, double lon) {
    EEPROM.put(0, lat);
    EEPROM.put(sizeof(double), lon);
    EEPROM.commit();
}

```

```

void loadLastLocation() {
    EEPROM.get(0, currentLat);
    EEPROM.get(sizeof(double), currentLon);
    Serial.printf("📦 Loaded last saved location: %.6f, %.6f\n", currentLat, currentLon);
}

```

// WIFI GEOLOCATION

```

void getWiFiLocation(double &lat, double &lon) {
    if (WiFi.status() != WL_CONNECTED) return;
    HTTPClient http;

```

```

http.begin("http://ip-api.com/json/");
http.addHeader("User-Agent", "ESP32Tracker");
int code = http.GET();
if (code == 200) {
    String payload = http.getString();
    StaticJsonDocument<512> doc;
    if (deserializeJson(doc, payload) == DeserializationError::Ok && doc["status"] == "success") {
        lat = doc["lat"].as<double>();
        lon = doc["lon"].as<double>();
        Serial.printf("📶 Wi-Fi Approx Location: %.6f, %.6f\n", lat, lon);
    } else {
        Serial.println("❌ JSON parse or location error.");
    }
} else {
    Serial.printf("❌ HTTP error: %d\n", code);
}
http.end();
}

```

CIRCUIT

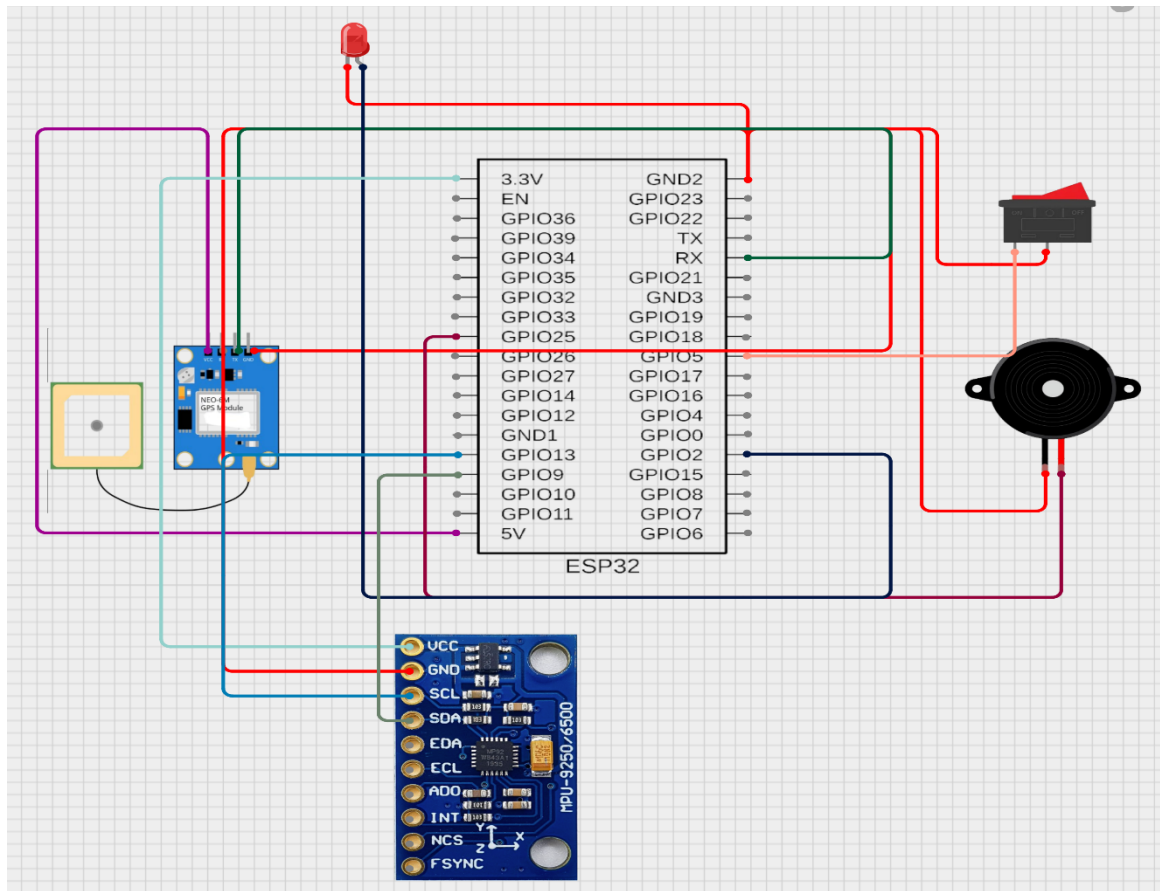


Figure-1: Circuit diagram showing connections between ESP32, GPS module, MPU6050, LED, buzzer, and switch.

WORKING OF DEVICE

- When the device is powered on, the ESP32 microcontroller initializes all connected components and starts communication with the GPS module and sensors.
- The GPS module (Neo-6M) begins receiving satellite signals and continuously updates the user's real-time location (latitude and longitude).
- The accelerometer (MPU6050) monitors motion and detects sudden movements, falls, or abnormal activity.
- In case of danger, the user can press the SOS button to trigger the emergency alert.
- If a sudden fall or violent movement is detected, the accelerometer automatically activates the alert system without manual input.
- Once an alert is triggered, the ESP32 collects the latest GPS coordinates from the GPS module.
- If the GPS module temporarily stops working or loses signal, the ESP32 still sends the emergency alert message using Wi-Fi, notifying contacts about the distress situation — even if location data is unavailable. A warning indicator (LED/Buzzer pattern) can signal GPS failure.
- The ESP32 transmits the alert (with or without GPS data) to predefined contacts or a cloud platform using Wi-Fi communication.
- The buzzer and LED indicator activate to give audio and visual confirmation that the alert has been successfully sent.
- The system continues to monitor the sensors and GPS module, automatically resuming location tracking once GPS connectivity is restored.
- The compact, low-power, and wearable design ensures that the user can carry the device easily and stay protected at all times.

OUTPUT

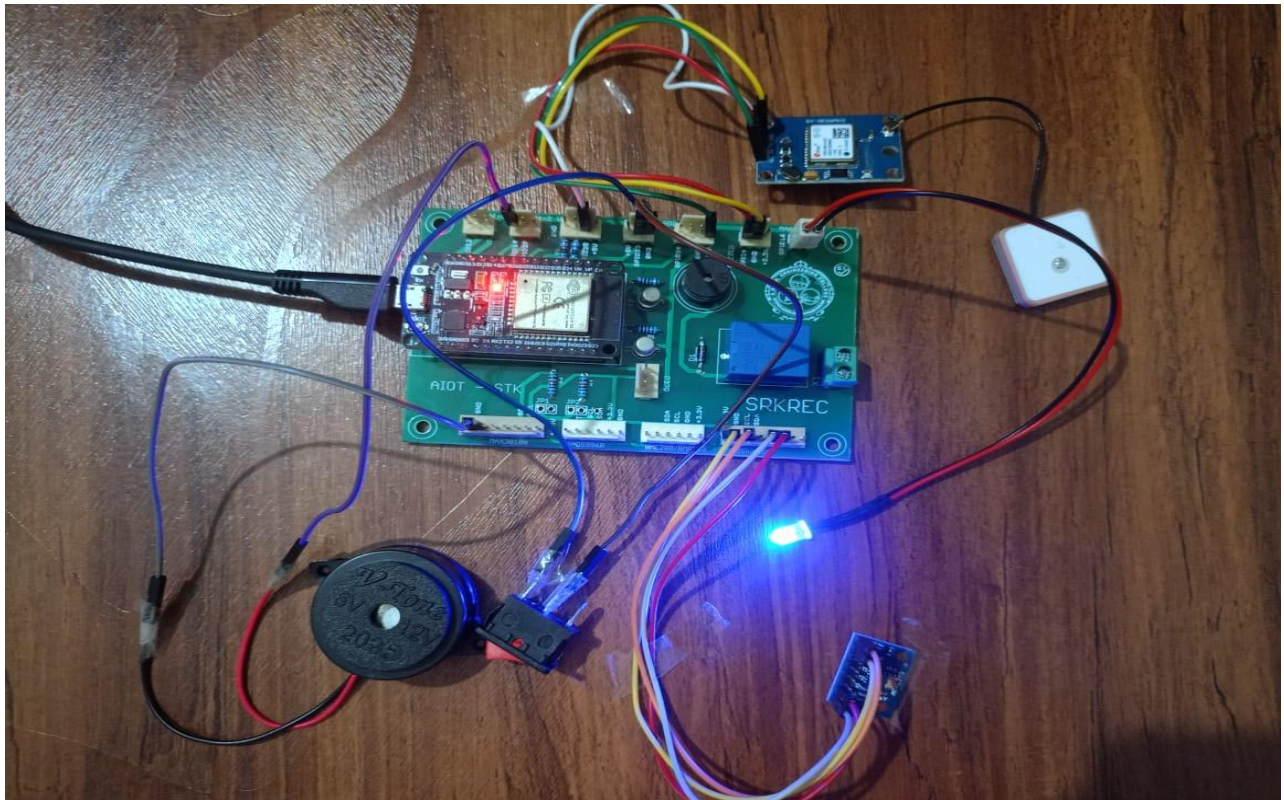


Figure -2 : Working of Esp32, GPS tracker

```
Output Serial Monitor X
Message (Enter to send message to 'ESP32 Dev Module' on 'COM6') Both NL & CR 115200 baud

config:ip: u, s,wp:uxee
clk_drv:0x00,q_drv:0x00,d_drv:0x00,cs0_drv:0x00,hd_drv:0x00,wp_drv:0x00
mode:DIO, clock div:1
load:0x3fff0030,len:4980
load:0x40078000,len:16612
load:0x40080400,len:3480
entry 0x400805b4

=====
Booting ESP32 GPS Tracker...
Loaded last saved location: 17.012899, 81.779297
Connecting to WiFi...
WiFi Connected!
IP Address: 192.168.10.35
Web Server Started Successfully!
System Ready. Press the switch to capture location.

SWITCH ACTIVATED - Capturing location...
GPS signal not available - using Wi-Fi location...
Wi-Fi Approx Location: 17.012899, 81.779297
Current Coordinates: 17.012899, 81.779297
Location saved to EEPROM memory.
Open your browser to view the map.

SWITCH ACTIVATED - Capturing location...
GPS signal not available - using Wi-Fi location...
Wi-Fi Approx Location: 17.012899, 81.779297
Current Coordinates: 17.012899, 81.779297
Location saved to EEPROM memory.
Open your browser to view the map.
```

Figure – 3 : Output of code on serial monitor

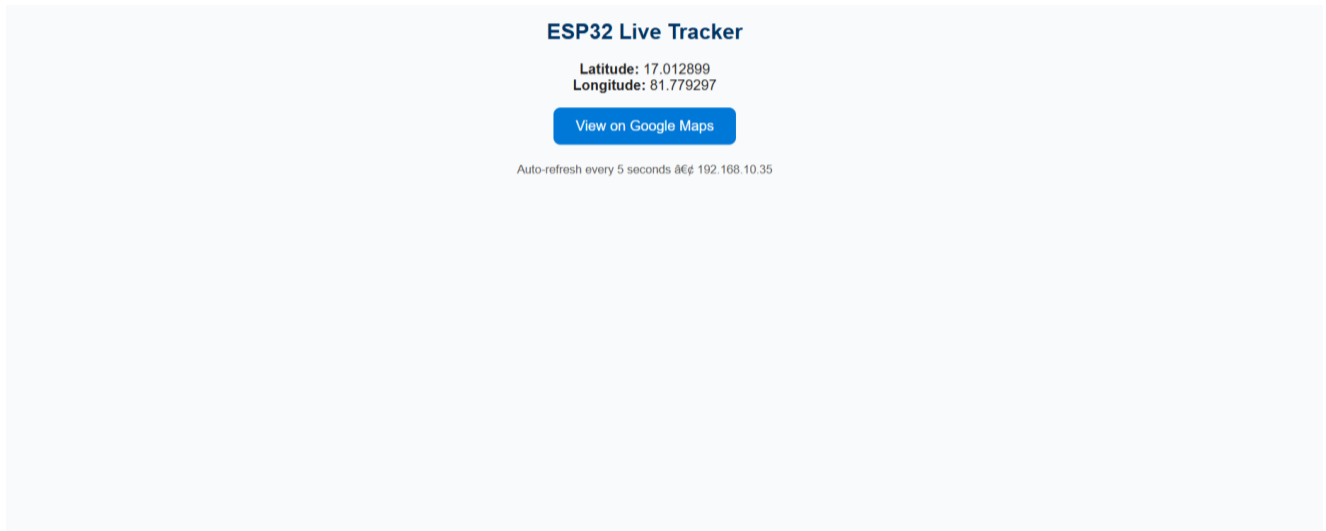


Figure – 4 : Website showing the location

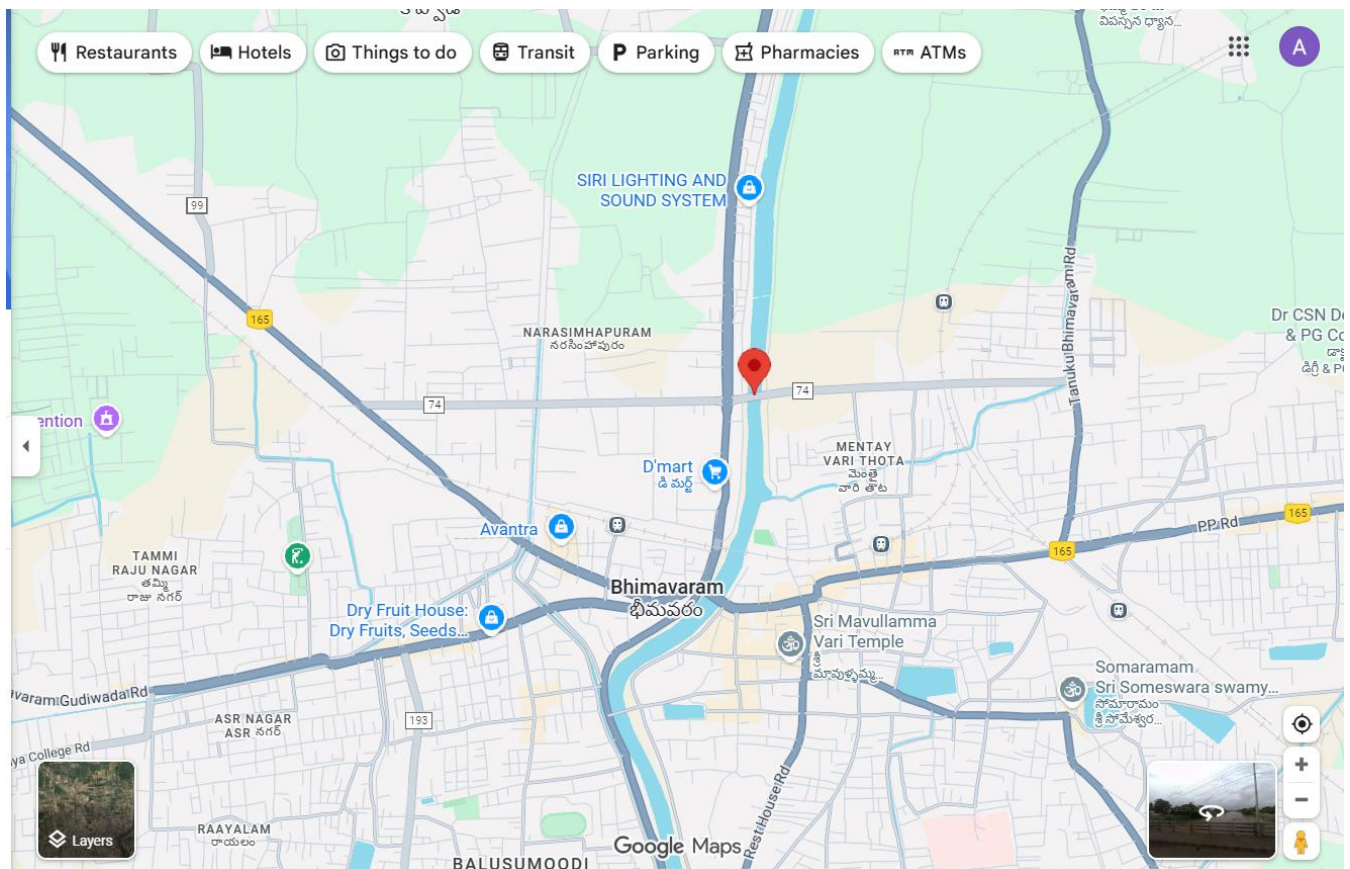


Figure – 5 : Location on google maps

CONCLUSION

The Women Safety Device using ESP32 is a smart and practical solution designed to enhance the safety of women in real-life situations. It continuously tracks the user's location using GPS and Wi-Fi, ensuring that the position can be identified even in areas with weak satellite signals. In case of an emergency, a single press of the switch activates both a visual and audible alert through the LED and buzzer while simultaneously updating the live location on a web page. This allows family members, friends, or authorities to respond quickly and accurately.

Beyond just being a technical project, this device reflects the growing importance of using IoT and embedded systems for social good. Its low cost, compact design, and ease of use make it suitable for practical implementation. Overall, this project demonstrates how technology can be a powerful tool in empowering women, improving personal safety, and promoting a sense of security in today's world.